034795

EXAMINATIONS COUNCIL OF ZAMBIA

Examination for School Certificate Ordinary Level

Mathematics 4024/2

Paper 2

Tuesday **7 NOVEMBER 2024**

Additional materials:
Answer Booklet
Silent Electronic Calculator (non programmable)
Geometrical instruments
Graph paper (3 sheets)
Plain paper (1 sheet)

Time: 2 hours 30 minutes **Marks: 100**

Instructions to Candidates

1 Write the **centre number** and your **examination number** on **every page** of the separate **Answer Booklet** provided.

2 Write your answers and working in the separate **Answer Booklet provided**.

3 If you use more than one Answer Booklet, fasten the Answer Booklets together.

4 Omission of essential working will result in loss of marks.

5 There are **twelve** questions in this paper.

(i) Section A
Answer **all** questions.

(ii) Section B
Answer any **four** questions.

6 **Silent non programmable Calculators may be used.**

Information for Candidates

1 The number of marks is given in brackets [ ] at the end of each question or part question.

2 If the degree of accuracy is not specified in the question, and if the answer is not exact, give the answer to three significant figures. Give answers in degrees to one decimal place.

3 Cell phones are **not allowed** in the examination room.

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Mathematical Formulae

1 ALGEBRA

Quadratic Equation

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

2 SERIES

Geometric Progression

$$S_n = \frac{a(1-r^n)}{1-r}, (r < 1)$$

$$S_n = \frac{a(r^n - 1)}{r-1}, (r > 1)$$

$$S\infty = \frac{a}{1-r} \text{ for } |r| < 1$$

3 TRIGONOMETRY

Formula for Δ ABC

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

$$a^2 = b^2 + c^2 - 2bc \cos A.$$

$$\Delta = \frac{1}{2} bc \sin A$$

4 STATISTICS

Mean and standard deviation

Ungrouped data

$$\text{Mean } (\bar{x}) = \frac{\Sigma x}{n}, \textbf{SD} = \sqrt{\left\{\frac{\Sigma(x-\bar{x})^2}{n}\right\}} = \sqrt{\left\{\frac{\Sigma x^2}{n} - (\bar{x})^2\right\}}$$

Grouped data

$$\text{Mean } (\bar{x}) = \frac{\Sigma fx}{\Sigma f}, \textbf{SD} = \sqrt{\left\{\frac{\Sigma f(x-\bar{x})^2}{\Sigma f}\right\}} = \sqrt{\left\{\frac{\Sigma fx^2}{\Sigma f} - (\bar{x})^2\right\}}$$

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SECTION A: (52 MARKS)

Answer all questions in this section.

1 **(a)** Simplify $\frac{15a^2b}{7cd^2} \div \frac{5a^3b^2}{21c^2d^3}$. [2]

(b) Solve the equation $2x^2 - x - 2 = 0$, giving your answers correct to 2 decimal places. [5]

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2 **(a)** Two boxes A and B contain balls of the same size. Box A contains 2 green balls and 3 blue balls while Box B contains 4 green balls and 2 blue balls. If one ball is selected at random from each box, what is the probability that both balls selected are

(i) blue, [2]

(ii) of different colours? [3]

(b) On a particular day at a school with 50 Grade 12 learners, 33 collected Biology textbooks, 24 collected Mathematics textbooks and 23 collected History textbooks from the school library. 13 learners collected both Biology and Mathematics textbooks, 16 collected both Mathematics and History textbooks, 15 collected both Biology and History textbooks and 10 collected all the three types of textbooks.

(i) Illustrate this information in a Venn diagram. [2]

(ii) How many learners collected

(a) none of the textbooks, [1]

(b) one type of textbook only, [1]

(c) two types of textbooks only? [1]

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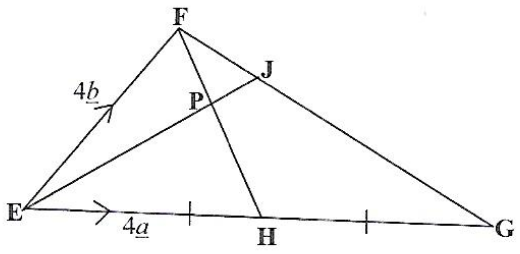
3 **(a)** The following program is written in the form of pseudocode.

```
Start
  Enter r, s
    IF s < r THEN
    Print 'error, s is not valid'
    ELSE A = π*r*s
    END IF
  Print A
Stop
```

Draw a flowchart corresponding to the pseudocode above. [5]

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(b) In the diagram, EFG is a triangle in which $\overrightarrow{EH} = 4\underline{a}$, $\overrightarrow{EF} = 4\underline{b}$, FJ : JG = 1 : 3 and H is the midpoint of EG. FH and EJ meet at P.



(i) Express in terms of $\underline{a}$ and/or $\underline{b}$

(a) $\overrightarrow{FH}$, [1]

(b) $\overrightarrow{FG}$, [1]

(c) $\overrightarrow{EJ}$. [1]

(ii) Given that $\overrightarrow{FP} = k\overrightarrow{FH}$, show that $\overrightarrow{EP} = 4k\underline{a} + 4(1-k)\underline{b}$. [2]

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4 **(a)** Given that matrix $P = \begin{pmatrix} 3 & -1 \\ -4 & 2 \end{pmatrix}$, find its

(i) determinant, [2]

(ii) inverse. [2]

(b) Given the geometric progression 10, 30, 90, …, find the

(i) 10th term, [2]

(ii) sum of the first 6 terms, [3]

(iii) geometric mean of 2 430 and 21 870. [2]

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5 **Answer the whole of this question on a sheet of plain paper.**

(a) Construct a quadrilateral PQRS in which PQ = 8cm, angle QPS = 110°, angle PQR = 88°, PS = 7cm and QR = 10cm. [1]

(b) Measure and write the length of RS. [1]

(c) Within the quadrilateral PQRS, construct the locus of points which are

(i) 6cm from QR, [1]

(ii) equidistant from PS and RS. [2]

(d) A point M, within the quadrilateral PQRS, is such that it is 6cm from QR and equidistant from PS and RS. Label the point M. [1]

(e) Another point N, within the quadrilateral PQRS, is such that it is less than or equal to 6cm from QR and nearer to PS than RS. Indicate, by shading, the region in which N must lie. [2]

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6 (a) Evaluate $\int_{-1}^{2} (4x^3 - 3x^2 + 2)\,dx$. [3]

(b) Express $\frac{1}{x-1} - \frac{10}{2x-1}$ as a single fraction in its simplest form. [3]

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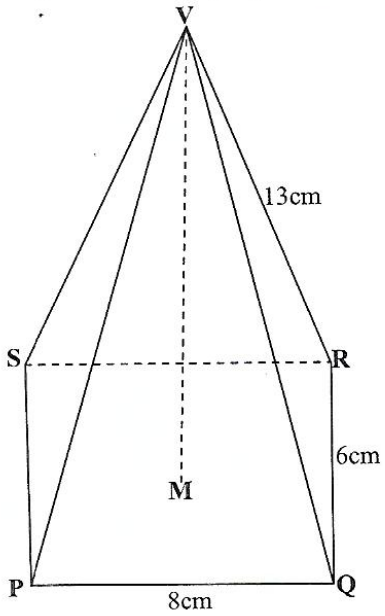
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SECTION B: [48 MARKS]

Answer any four questions in this section.

Each question in this section carries 12 marks.

7 **(a)** The diagram is a rectangular based right pyramid.

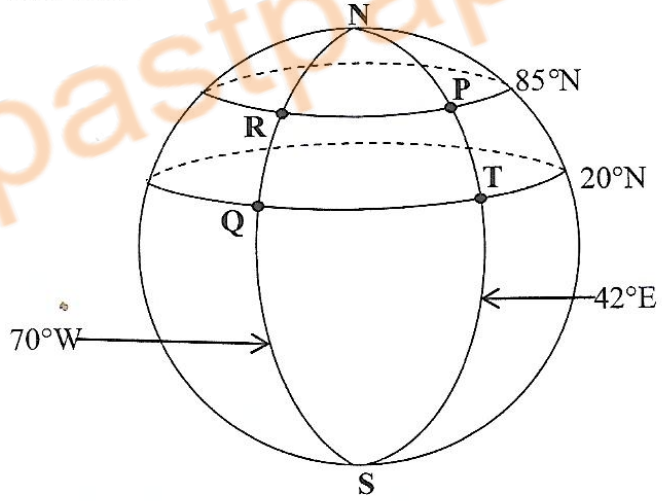


Given that PQ = 8cm, QR = 6cm and SV = PV = QV = RV = 13cm, calculate the

(i) perpendicular height, MV, of the pyramid, [3]

(ii) volume of the pyramid. [3]

(b) The diagram shows the points R, Q, P and T on the surface of the earth.
[Take $\pi = 3.142$ and R = 3 437nm]



(i) Find the difference in longitudes between points Q and T. [2]

(ii) Calculate, in nautical miles, the distance between

(a) R and P along latitude 85°N, [2]

(b) P and T along longitude 42°E. [2]

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8 The table shows the time, in minutes, taken by 80 learners to complete a task in a Science experiment.

Time (minutes)	$0 < x \leq 2$	$2 < x \leq 4$	$4 < x \leq 6$	$6 < x \leq 8$	$8 < x \leq 10$	$10 < x \leq 12$
Number of learners	5	15	25	20	10	5

(a) Calculate the standard deviation. [6]

(b) **Answer this part of the question on a sheet of graph paper.**

(i) Using the table above, copy and complete the cumulative frequency table below.

Time (minutes)	≤ 0	≤ 2	≤ 4	≤ 6	≤ 8	≤ 10	≤ 12
Number of learners	0	5	20	45			

[1]

(ii) Using a scale of 2cm to represent 2 units on the x-axis for $0 \leq x \leq 12$ and 2cm to represent 10 units on the y-axis for $0 \leq y \leq 80$, draw a smooth cumulative frequency curve. [3]

(iii) Showing your method clearly, use your graph to estimate the semi-interquartile range. [2]

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9 **Answer the whole of this question on a sheet of graph paper.**

A businessman intends to make two types of products, A and B for sale. The cost of making product A is K60.00 and the cost of making product B is K30.00. The businessman has K3 000.00 to spend on the products. He decides to produce at least 20 of product A and at least 10 of product B.

(a) Let x be the number of product A and y be the number of product B. Write three inequalities which represent these conditions. [4]

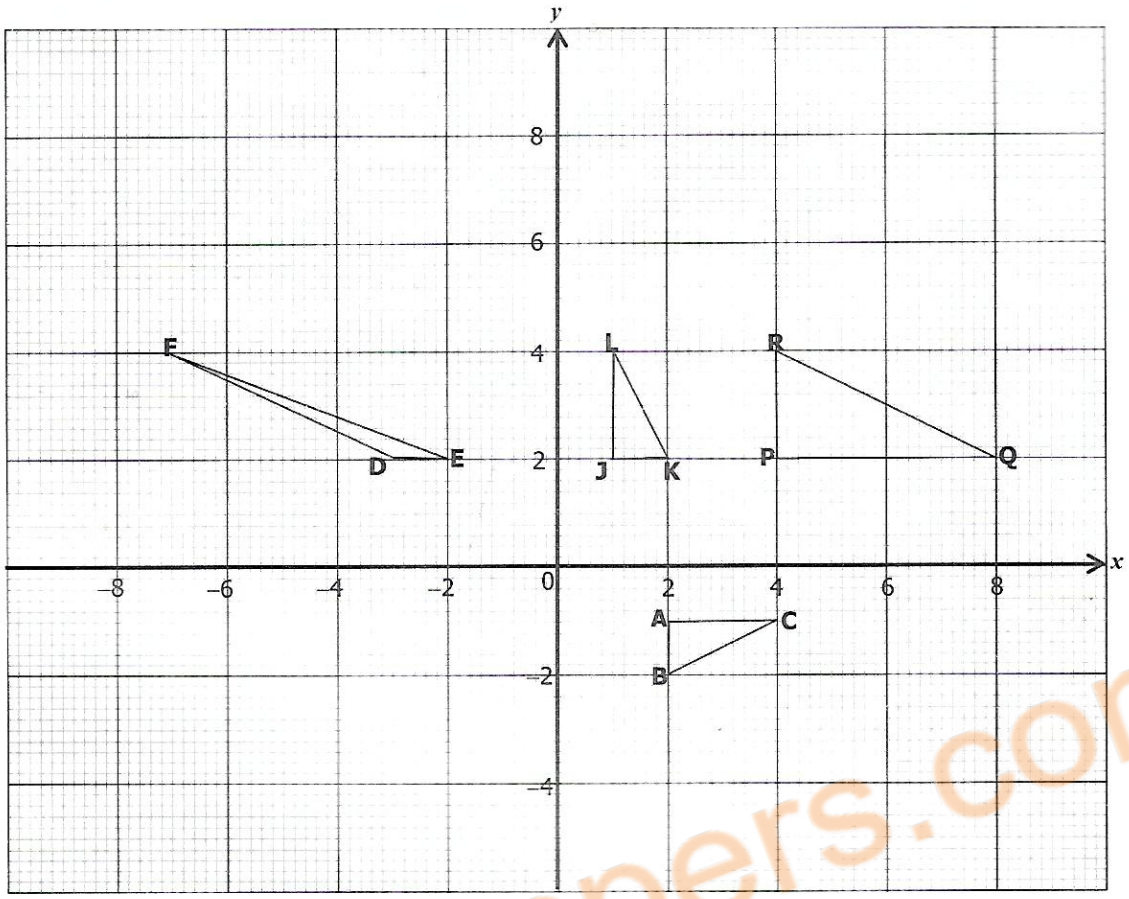
(b) Using a scale of 2cm to represent 10 units on each axis, draw the x-axis for $0 \leq x \leq 60$ and y-axis for $0 \leq y \leq 100$ respectively and shade the unwanted region to show clearly the region in which the solution of the inequalities lie. [4]

(c) The profit on the sale of product A is K80.00 and profit on the sale of product B is K50.00. How many products of each type can he make to maximise profit? [2]

(d) Calculate the maximum profit. [2]

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10 Study the following diagram and answer the questions that follow.



(a) A clockwise rotation maps triangle JKL onto triangle ABC. Find the centre and angle of this transformation. [2]

(b) A transformation with matrix $\begin{pmatrix} -2 & 0 \\ 0 & -2 \end{pmatrix}$ maps triangle JKL onto triangle XYZ, not shown on the diagram. Find the

(i) scale factor of this transformation, [1]

(ii) coordinates of X, Y and Z. [3]

(c) Triangle JKL is mapped onto triangle PQR by a stretch. Find the matrix which represents this transformation. [3]

(d) Describe fully a single transformation which maps triangle JKL onto triangle DEF. [3]

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11 (a) Answer this part of the question on a sheet of graph paper.

The values of x and y are connected by the equation $y = x^3 - 6x + 2$. Some corresponding values of x and y are given in the table below.

x	−3	−2	−1	0	1	2	3
y	−7	6	7	2	−3	−2	k

(i) Calculate the value of k. [1]

(ii) Using a scale of 2cm to represent 1 unit on the x-axis for $-3 \leq x \leq 3$ and 2cm to represent 5 units on the y-axis for $-10 \leq y \leq 15$, draw the graph of $y = x^3 - 6x + 2$. [3]

(iii) Use your graph to solve the equations

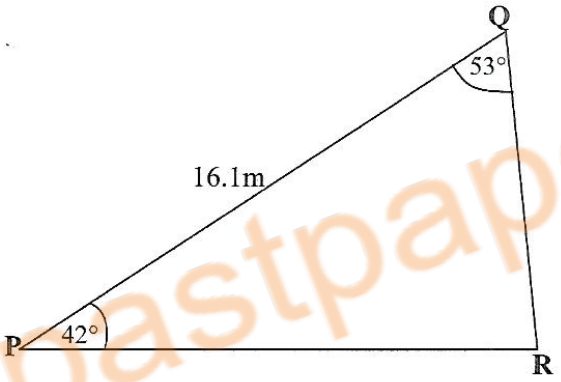
(a) $x^3 - 6x + 2 = 0$, [2]

(b) $x^3 - 6x = 2$. [3]

(b) Find the equation of the normal to the curve $y = x^3 + 3x^2 + 2$ at the point $(-1, 4)$. [3]

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12 (a) In triangle PQR, PQ = 16.1m, angle QPR = 42° and angle PQR = 53°.



Calculate the

(i) length of QR, [4]

(ii) area of triangle PQR, [2]

(iii) shortest distance from R to PQ. [2]

(b) Solve the equation $6 \sin \theta = -3$, for $180° \leq \theta \leq 360°$. [2]

(c) Simplify $\frac{x^3 - 9x}{x + 3}$. [2]

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